

Computer Networks

Assignment No. 3

Question No. 1

Admin (60 hosts)

- Hosts (h) = $2^6 - 2 = 62$, $h = 6$
- Subnet Mask = $32 - 6 = /26$
- Network address : 192.168.50.0
- 1st valid Host : 192.168.50.1
- Last valid Host : 192.168.50.62
- Broadcast address : 192.168.50.63

HR Department (30 hosts)

- Hosts (h) = $2^5 - 2 = 30$, $h = 5$
- Subnet Mask = $32 - 5 = /27$
- Network Address : 192.168.50.64
- 1st valid Host : 192.168.50.65
- Last valid Host : 192.168.50.94
- Broadcast address : 192.168.50.95

Sales Dept (28 hosts)

- Hosts (h) = $2^5 - 2 = 30$; h = 5
- Subnet Mask = $32 - 5 = /27$
- Network address : 192.168.50.96
- 1st valid Host : 192.168.50.97
- Last valid Host : 192.168.50.126
- Broadcast address : 192.168.50.127

IT Dept (12 hosts)

- Hosts (h) = $2^4 - 2 = 14$; h = 4
- Subnet Mask = $32 - 4 = /28$
- Network address = 192.168.50.128
- 1st valid address = 192.168.50.129
- Last valid address = 192.168.50.142
- Broadcast address = 192.168.50.143

Security Dept (6 Hosts)

- Hosts (h) = $2^3 = 8$; h = 3
- Subnet Mask = $32 - 3 = /29$
- Network address = 192.168.50.144
- 1st valid host = 192.168.50.145
- last valid host = 192.168.50.150
- Broadcast address = 192.168.50.151

Question No. 2

10.10.0.0/22

$$2^2 = 4$$

Adding 2 bits to original CIDR = /24

Head office

- Network Address : 10.10.0.0
- CIDR Notation : /24
- Broadcast address : 10.10.0.255
- 1st Usable IP : 10.10.0.1
- Last Usable IP : 10.10.0.254

Lahore

- Network Address : 10.10.1.0
- CIDR Notation : /24
- Broadcast address : 10.10.1.255
- 1st Usable IP : 10.10.1.1
- Last Usable IP : 10.10.1.254

Karachi

- ⇒ Network Address : 10.10.20
- ⇒ CIDR Notation : . . . /24
- ⇒ Broadcast address: 10.10.2.255
- ⇒ 1st Usable IP : 10.10.2.1
- ⇒ last Usable IP : 10.10.2.254

Islamabad

- ⇒ Network Address : 10.10.3.0
- ⇒ CIDR Notation : /24
- ⇒ Broadcast address: 10.10.3.255
- ⇒ 1st Usable IP : 10.10.3.1
- ⇒ last Usable IP : 10.10.3.254